Two orchids: reproducing an adapt3 community projection

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```
ENV["GKSwstype"] = "100"
using MatrixCommunityModels
using Plots
```

The model

The R package adapt3 (Shefferson 2026) projects several matrix population models against a common density. Its community-projection vignette uses two lady's slipper orchids, *Cypripedium candidum* and *C. parviflorum*, each with nine stages (dormant seed `SD`, protocorm `P1`, seedling `SL`, dormant adult `D`, and five sizes of flowering adult) and one matrix per study year, estimated from field data with the R package lefko3.

In adapt3 the community is specified by four parallel lists: matrices, start vectors, density frames keyed by matrix element, and equivalence frames. Here the same information is three statements about each species:

- its transitions cycle through the annual matrices (the environment of step `t` is the year counter);
- reproduction into `SD` and `P1` is multiplied by Ricker terms, `0.5 exp(-N)` and `1.2 exp(-2N)`, where `N` is the total of the `density` pool;
- every stage except dormant seed occupies the `density` pool with weight one (adapt3's equivalence frame).

```
m = cypripedium_model()
m.community
```

```
OpenCommunity(candidum, parviflorum; pools = density)
```

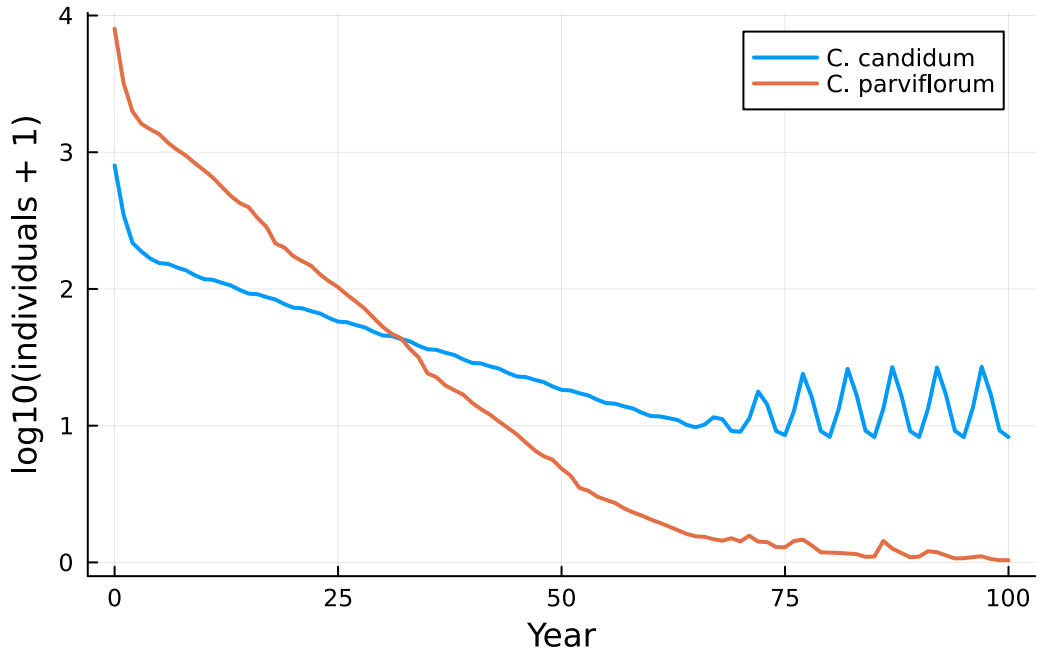
The estimated matrices are shipped as data (`data/cypripedium`), since estimating them needs the R workflow. The transitions that read the pool are the ones into `SD` and `P1` from flowering stages:

```
sp = species(m.community, :candidum)
[(t.name, t.kind) for t in sp.transitions if t.to in (:SD, :P1) && t.from
 ↪  in (:XSm, :Sm, :Md, :Lg, :XLg)]
```

```
10-element Vector{Tuple{Symbol, Symbol}}:
 (:XSm_to_SD, :fecundity)
 (:XSm_to_P1, :survival)
 (:Sm_to_SD, :fecundity)
 (:Sm_to_P1, :survival)
 (:Md_to_SD, :fecundity)
 (:Md_to_P1, :survival)
 (:Lg_to_SD, :fecundity)
 (:Lg_to_P1, :survival)
 (:XLg_to_SD, :fecundity)
 (:XLg_to_P1, :survival)
```

Projection

```
sol = project(m.community, m.n0, m.envs)
totals = [sum(sol[:candidum]; dims = 2) sum(sol[:parviflorum]; dims = 2)]
plot(sol.t, log10.(totals .+ 1); label = ["C. candidum" "C. parviflorum"],
 ↪  xlabel = "Year",
     ylabel = "log10(individuals + 1)", lw = 2)
```



With these Ricker terms recruitment is almost completely suppressed while any appreciable number of plants is present (`exp(-N)` with `N` in the hundreds), so both populations decline, and *C. parviflorum* shows brief bursts of recruitment when the community becomes very small.

Agreement with adapt3

The reference files were produced by `project3()` in adapt3 2.0.9 from the vignette's own objects (`test/reference/R/run_adapt3_reference.R`). They hold the size of each population and the aggregate density at each time.

```
function reference(tag)
    file = joinpath(pkgdir(MatrixCommunityModels), "test", "reference",
 ↪  "adapt3_$(tag)_N.csv")
    permutedims(reduce(hcat, [parse.(Float64, split(l, ",")[2:4]) for l in
     ↪  readlines(file)[2:end]]))
end
ref = reference("continuous")
(max_scaled_difference = maximum(abs.(totals .- ref[:, 1:2]) ./ max.(ref[:,
 ↪  1:2], 1.0)),
 density_matches = sol.pools[1:100, 1] ≈ ref[1:100, 3])
```

```
(max_scaled_difference = 8.268022275869569e-16, density_matches = true)
```

The vignette itself floors the stage vectors to whole individuals each step (`integeronly = TRUE`). That is a threshold applied after the projection:

```
mi = cypripedium_model(integeronly = true)
soli = project(mi.community, mi.n0, mi.envs)
totalsi = [sum(soli[:candidum]; dims = 2) sum(soli[:parviflorum]; dims = 2)]
totalsi == reference("vignette")[:, 1:2]
```

```
true
```

```
Int.(totalsi[1:12, :])'
```

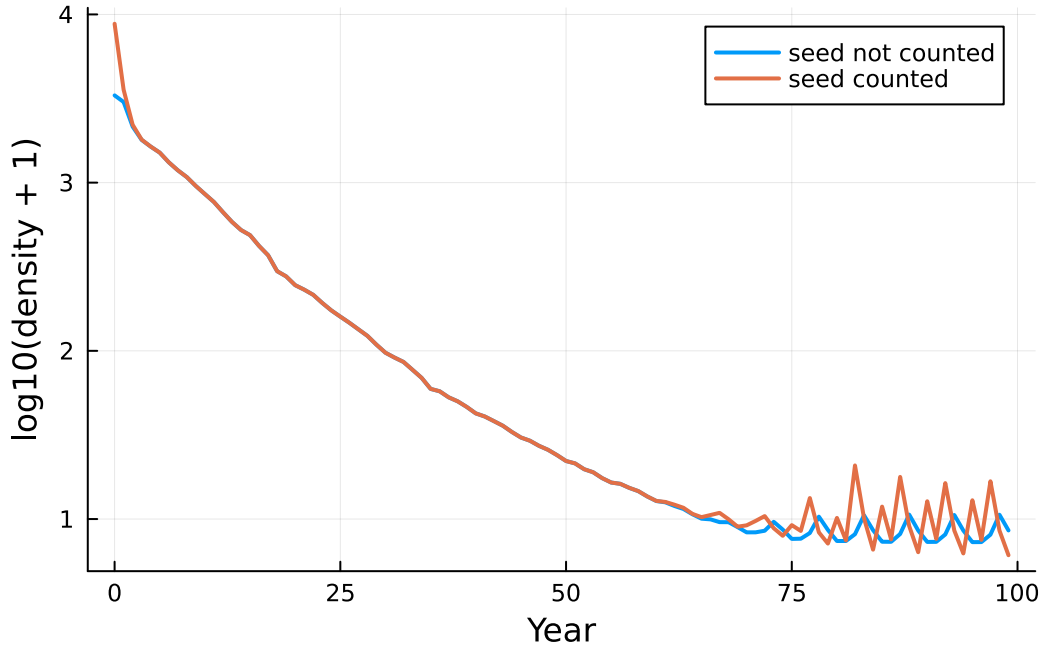
```
2×12 adjoint(::Matrix{Int64}) with eltype Int64:
  800   347   213   180   157   143   138   128  120  108  100   95
 8000  3215  1987  1608  1464  1344  1158  1028  928  815  723  635
```

These are the numbers printed in the adapt3 vignette. *C. candidum* is extinct from year 30 and *C. parviflorum* from year 69.

Changing the dependency assumption

Because the weights are part of each species, the effect of an assumption can be seen by changing it in one place. If dormant seed counted toward density:

```
ma = cypripedium_model(equivalence = false)
sola = project(ma.community, ma.n0, ma.envs)
plot(sol.t[1:end-1], log10.([sol.pools[1:end-1, 1] sola.pools[1:end-1, 1]]
 ↪  .+ 1);
     label = ["seed not counted" "seed counted"], xlabel = "Year", ylabel =
 ↪  "log10(density + 1)", lw = 2)
```



Reference

Shefferson, R. P. (2026). adapt3: Adaptive dynamics and community projection in the R programming language. *Methods in Ecology and Evolution* 17, 1719-1725. https://doi.org/10.1111/2041-210x.70298